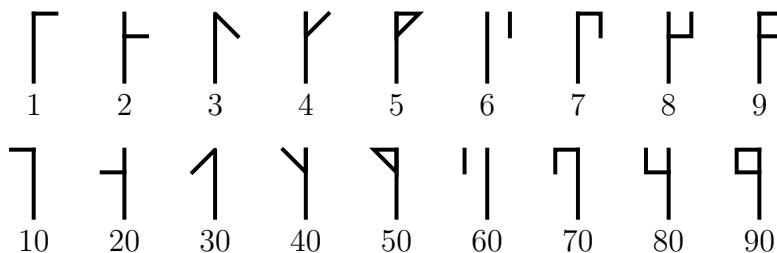


GRADES 5 AND 6

SAMPLE QUESTION FOR 3 POINTS

Cistercian numerals were used in the early thirteenth century. Any integer from 1 to 99 can be represented by a single glyph formed by combining two of the glyphs shown below.

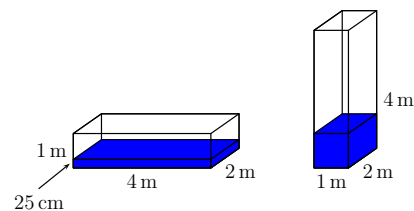


The glyph for 24 looks like , the glyph for 81 looks like , and the glyph for 93 looks like . What does the glyph for 45 look like?

- (A)  (B)  (C)  (D)  (E) 

SAMPLE QUESTION FOR 4 POINTS

A water tank with a rectangular base has the dimensions $1\text{ m} \times 2\text{ m} \times 4\text{ m}$. It contains water to a depth of 25 cm, as shown in the first picture. The tank is turned so that a $1\text{ m} \times 2\text{ m}$ face becomes the base, as shown in the second picture. What is the depth of the water now?



- (A) 25 cm (B) 50 cm (C) 75 cm (D) 1 m (E) 1.25 m

SAMPLE QUESTION FOR 5 POINTS

Some glasses are stacked on top of each other. A stack of 8 glasses is 42 cm high and a stack of 2 glasses is 18 cm high. How high is a stack of 6 glasses?

- (A) 22 cm (B) 24 cm (C) 28 cm (D) 34 cm (E) 40 cm



GRADES 7 AND 8

SAMPLE QUESTION FOR 3 POINTS

The license plate of Kangy's car fell off. He put it back upside down but luckily this didn't make any difference. Which of the following could be Kangy's license plate?

- (A) **04 NSN 40** (B) **60 HOH 09** (C) **80 BNB 08**
 (D) **03 HNH 30** (E) **08 XBX 80**

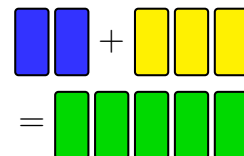
SAMPLE QUESTION FOR 4 POINTS

In my office, there are two clocks. One clock gains one minute every hour and the other loses two minutes every hour. Yesterday I set them both to the correct time but when I looked at them today, I saw that the time shown on one was 11:00 a.m. and shown on the other was 12:00 noon. What time was it when I set the two clocks?

- (A) 11:00 p.m. (B) 7:40 p.m. (C) 3:40 p.m. (D) 2:00 p.m. (E) 11:20 a.m.

SAMPLE QUESTION FOR 5 POINTS

A painter wanted to mix 2 liters of blue paint with 3 liters of yellow paint to make 5 liters of green paint. However, by mistake he used 3 liters of blue and 2 liters of yellow so that he made the wrong shade of green. What is the smallest amount of this green paint that he must throw away so that, using the rest of his green paint and some extra blue and/or yellow paint, he could make 5 liters of paint of the correct shade of green?



- (A) $\frac{5}{3}$ liters (B) $\frac{3}{2}$ liters (C) $\frac{2}{3}$ liters (D) $\frac{3}{5}$ liters (E) $\frac{5}{9}$ liters